MAXM86146EVSYS Compatibility Matrix

MAX32664C for MAXM86146 .msbl version	MAXM86146 GUI Version	NRF52832 version
33.6.0 (MAX86146EVSYS is factory programmed to 33.6.0)	1.0.0 (MAXM86146SensorHubGUISetupV100.exe)	0.1 (MAX86146EVSYS is factory programmed to 0.1)
33.12.0	1.0.0 (MAXM86146SensorHubGUISetupV100.exe)	0.1 (MAX86146EVSYS is factory programmed)
33.13.31	1.0.0 (MAXM86146SensorHubGUISetupV100.exe)	0.1 (MAX86146EVSYS is factory programmed)

Sensor Hub .msbl Version Numbering Convention

Integrated Sensor Hub + Optical Sensor	.msbl versions	Form Factor
MAXM86146CFU+	MAX32664C_OB07_WHRM_AEC_SCD_WSPO2_C_33.x.y.msbl (MAX86146EVSYS Website)	Chest

MAX32664C 33.x.x .msbl for MAXM86146 Firmware Release Notes

Version	Date	Importance	History	Known Issues
33.13.31	4/19/22	Medium	- Normal algorithm data output FIFO samples limited to 20 bytes - Extended algorithm data output FIFO samples limited to 52 bytes - Deep sleep while idle restored. - Parts of the .msbl firmware compiler optimized for size to fit latest algorithm. - Wearable Algorithm Suite 8.4.0 ○ HR and SpO2 Performance Enhancements ○ SpO2 algorithm is updated to reject negative PPG ○ AEC algorithm is updated to reject negative PPG ○ Improved sudden SpO2 drop detection. ○ Improved SpO2 Confidence Level ○ WHRM Motion Frequency usage is enhanced to increase HRM accuracy - AEF Controller Enhancement ○ Logic Improvement for SNR calculation when LED current cannot reach calculated target ○ Re-trigger SNR calculation if optical contact deteriorates - I2C command 0x20 added as a synonym to 0x12 00 - I2C command 0x21 added as a synonym to 0x12 01 - I2C command 0x52 0x07 0x04 added, packed normal samples report - I2C command 0x50 0x07 0x20 added, config mask for packed -samples fifo data - I2C command 0x50 0x07 0x20 added, config mask for packed -samples fifo data - I2C command 0x01 0x00 0x08 added, combined with RSTN, BL mode is entered	- ST accel is high resolution mode; low resolution mode was too noisy to be useful. - SCDSM not supported for ST LIS2DS12 accelerometer part - Initial HR setting configuration is not operational. - 0x11 0x03, read I2C address command is not supported. - When using the MAX86146EVSYS, wait 5 seconds between the flashing the Nordic .hex and the MAX32664 .msbl, cycle power to the device and wait 5 seconds between connecting and clicking on the Start button.

33.13.19 (engineering release)	5/19/19	High	- Wearable Algorithm Suite v6.5.4 - Add configurable initial integration time (0x50 0x07 0x1A) - Add configurable initial sampling rate, averaging (0x50 0x07 0x1B) - Add configurable SpO2 Red-IR PI threshold (0x50 0x07 0x1D) - IBI offset calculations updated to 1 sec average for Wellness algorithm. - 1 reserve byte in algo output changed to SpO2 orientation flag for Wellness Sleep algorithm. - Fix KX122 detection - Add KX112 as a supported accel (not tested) - Add read algo output size (0x11 0x06 0x01/02/03).	- ST accel is high resolution mode; low resolution mode was too noisy to be useful. - SCDSM not supported for ST LIS2DS12 accelerometer part - Initial HR setting configuration is not operational. - 0x11 0x03, read I2C address command is not supported. - Not compatible with MAXM86146 PC GUI
33.13.12	3/11/21	High	- Wearable Algorithm Suite 5.0 - Four bytes added to the Algo Ouput FIFO - 1Hz IBI reporting mode - Sensor hub automatically supports either STM LIS2DS12 or KX122. - ST accel to high resolution mode - Algo Hub mode	- ST accel is high resolution mode; low resolution mode was too noisy to be useful. - SCDSM not supported for ST LIS2DS12 accelerometer part - Initial HR setting configuration is not operational. - 0x10 0x00 0x08, enter bootloader mode is not functional. - 0x11 0x03, read I2C address command is not supported. - KX122 not detected.
33.13.4	1/18/21	Medium	- Fix LED, PD, Slot configuration commands: 0x50 0x07 0x17/0x18/0x19 - Four bytes added to the algorithm output FIFO report (IBI offset, and three reserved bytes) - Updated to support STM LIS2DS12 for the sensor hub accel - SCD only mode added. - SCD state machine for MFIO interrupt mode (SCDSM) added.	- MAX32630 sample host code has not been updated to handle the extra four bytes added in the algorithm output FIFO report.
33.12.0	7/22/20	High	- Wearable Algorithm Suite 2.10.0 - Improved SpO2 performance - Event reporting over mfio pin (MFIO mode) - Skin Contact State Machine in MFIO mode - fast converging Spo2 - fast converging to resting hearth rate - Enhanced MLP model - Command added to change LED firing (0x07 0x19) - Definition of command (0x07 17 and 0x07 18) are changed. They specify Slot # rather than LED #. THIS MAY CAUSE BACKWARD COMPATIBILITY IF NON-DEFAULT LED/PD CONFIGURATION IS USED - Peak detector module for IBI detection enhanced to reduce false detection rate - MLP integration to improve HR accuracy - Fixed bug of skin detection when heart rate measurement is done on Red or Red PPG signals - Fixed bug of lagging PPG samples after long term run - Improved authentication commands - Motion frequency tracking for HR measurement and skin contact detection are improved - Algorithm configuration command to set initial value of sensor sample rate/average and time integration - Fixed bug of sampled HR mode - Authentication feature and commands - Automatically detects number of PDs supported by MAX86140 or MAX86141 sensors - Motion frequency tracking for HR measurement improved - Bug fix in poll period > 1min causing overflow in I2C driver	- When using the MAX86146EVSYS, wait 5 seconds between the After flashing the Nordic .hex and the MAX32664 .msbl, cycle power to the device and wait 5 seconds between connecting and clicking on the Start button.

			- Reported number of samples in output FIFO (Family byte=0x12, index=0x00) changed from 1 to two byte, LSB first - Sensor Hub status [7:0] bit 6 is used to flag for Host accel underflow if samples are not feed fast enough to sensor hub - Sensor Hub accelerometer sampling freq changed from 100Hz to 200Hz to support 100Hz raw streaming. - HR RR and RR confidence are reported only when calculated (typically once per several samples) and zero the rest of the time. - Initials sampling frequency of sensor set to 100Hz with avg=4. - Default LED PD Configuration changed to suit MRD103: - WHRM: from 0x0073 to 0x0001 (both PDs are feed to WHRM algorithm) - SPO2: from 0x1121 to 0x1020 (Uses far PD1 for red/Ir in Spo2).	
33.6.0	10/10/19	High	- LED firing sequence configuration (index 0x19) for 6 slots Default: LED1, LED2, LED3, LED4, LED5, LED6 - PD and LED configuration (index 0x17 and 0x18), Default: SLOT1/PD1 for WHRM, LED3/2 for ir/red and PD2 for SPO2 - All features from MAX32664C v.30.6.0	- SCD based power saving support is not functional. - When using the MAX86146EVSYS, wait 5 seconds between the After flashing the Nordic .hex and the MAX32664 .msbl, cycle power to the device and wait 5 seconds between connecting and clicking on the Start button.